

stat_cate_levelset_supermargin

Specialization: v1

TL;DR. For the fixed calibrated constants, the nonempty super-margin frontier is exactly $\{(1, \gamma) : \gamma > d\}$, and an explicit transverse law proves nonemptiness with an interior threshold crossing. The minimax risk is exactly of order $(r_n^*)^2$ when $\beta \geq \gamma$. For $\beta < \gamma$, the full two-branch match remains open: the published low-smoothness fuzzy experiment uses covariate-density holes and therefore does not belong to the calibrated class, whose density is uniform on the entire cube.

1 Project justification

Gap. Bonvini–Kennedy–Keele give the conditional upper order $(r_n^*)^{1+\xi}$, while their lower theorem excludes $\xi\gamma > d$. Kennedy–Balakrishnan–Robins–Wasserman obtain the low-smoothness CATE scale by a fuzzy construction whose covariate density omits bump-transition regions. That construction cannot be inserted into the present calibrated class because calibration forces a positive uniform density on every such region.

Niche. The note isolates a calibration-specific, nonempty transverse super-margin class with an exact two-sided margin and weighted level-set loss. It separates the part of the statistical problem settled by a direct graph Assouad construction from the unresolved known-uniform-design nuisance-sensitive converse.

Fill. The proved contribution is the exact calibrated feasibility frontier, an explicit causal witness with interior crossing, well-posed weighted risk, and an honest minimax bracket that matches $(r_n^*)^2$ for $\beta \geq \gamma$. The residual result is sharpened to a concrete obstruction: the standard fuzzy lower experiment loses mixture matching once uniform mass is restored on smooth transition regions.

2 Related work

Bonvini–Kennedy–Keele [1] define the weighted CATE level-set loss and prove a conditional upper bound of order $(r_n^*)^{1+\xi}$, but their minimax lower theorem assumes $\xi\gamma \leq d$. Kennedy–Balakrishnan–Robins–Wasserman [2] derive the two-branch pointwise CATE scale using localized fuzzy hypotheses. Their low-smoothness construction deliberately supports the covariate law on flat bump tops so that the averaged one-observation densities coincide; restoring full uniform support creates a first-order unmatched-mixture term and recovers only the oracle converse scale. Audibert–Tsybakov [3] supplies the adjacent Hölder-margin crossing obstruction, while Rigollet–Vert [4] is a noncausal level-set analogue rather than a solution to this uniform-design causal converse.

3 Setup

Target (estimand / structural parameter): $G_P = \{x \in \mathcal{X} : \tau_P(x) > \theta\}$. Primary functional / estimator / diagnostic: $\mathcal{F}(\kappa) = \{(1, \gamma) : \gamma > d\}$. For every fixed calibrated κ , every $\gamma > d$, and all sufficiently large n ,

$$c_{\text{lb}} n^{-2\gamma/(2\gamma+d)} \leq \mathfrak{R}_n(1, \gamma; \kappa) \leq C_{\text{ub}} n^{-2 \min\{\beta, \gamma\} / \{2 \min\{\beta, \gamma\} + d\}}$$

. This equals $(r_n^*)^2$ in order when $\beta \geq \gamma$; the exact rate for $\beta < \gamma$ remains open..

- n — integer; $\mathbb{N}$; asymptotic index

Definition. sample size

- d — integer; $\mathbb{N}$; dimension

Definition. covariate dimension

- c — scalar; $(0, 1/4)$; calibration constant

Definition. calibration slope

- $\mathcal{X}$ — set; $\mathbb{R}^d$; support
Definition. fixed covariate support
- M_Y — scalar; $(0, \infty)$; model constant
Definition. outcome envelope
- θ — scalar; $\mathbb{R}$; decision threshold
Definition. fixed CATE threshold
- η — scalar; $(0, \infty)$; localization constant
Definition. threshold-band width
- t_0 — scalar; $(0, \infty)$; margin constant
Definition. margin range endpoint
- c_f — scalar; $(0, \infty)$; model constant
Definition. density lower constant
- C_f — scalar; $(0, \infty)$; model constant
Definition. density upper constant
- c_π — scalar; $(0, 1/2)$; model constant
Definition. overlap constant
- c_m — scalar; $(0, \infty)$; model constant
Definition. margin lower constant
- C_m — scalar; $(0, \infty)$; model constant
Definition. margin upper constant
- L_π — scalar; $(0, \infty)$; model constant
Definition. propensity Hölder radius
- L_0 — scalar; $(0, \infty)$; model constant
Definition. control-regression Hölder radius
- L_τ — scalar; $(0, \infty)$; model constant
Definition. CATE Hölder radius
- α — scalar; $(0, \infty)$; smoothness index
Definition. propensity Hölder order
- β — scalar; $(0, \infty)$; smoothness index
Definition. control-regression Hölder order
- γ — scalar; $(0, \infty)$; smoothness index
Definition. near-band CATE Hölder order
- γ' — scalar; $(0, \infty)$; smoothness index
Definition. off-band CATE Hölder order
- ξ — scalar; $(0, \infty)$; margin index
Definition. tight-margin exponent

- κ — parameter vector; $2^{\{\mathbb{R}^d\} \times (0, \infty) \times \mathbb{R} \times (0, \infty)^2 \times (0, \infty)^2 \times (0, 1/2) \times (0, \infty)^5 \times (0, \infty)^2 \times (0, \infty)}$; model index

Definition. $\kappa = (\mathcal{X}, M_Y, \theta, \eta, t_0, c_f, C_f, c_\pi, c_m, C_m, L_\pi, L_0, L_\tau, \alpha, \beta, \gamma')$

- $\bar{s}$ — scalar; $(0, \infty)$; rate index

Definition. average nuisance smoothness

- ρ — scalar; $(0, \infty)$; function-class index

Definition. generic Hölder order

- L — scalar; $(0, \infty)$; function-class index

Definition. generic Hölder radius

- U — set; $2^{\{\mathbb{R}^d\}}$; function-class index

Definition. generic Hölder domain

- $\mathcal{H}_{\rho}(L; U)$ — function class; $\{g: U \rightarrow \mathbb{R}\}$; smoothness class

Definition. Taylor-remainder Hölder ball

- x — deterministic point; $\mathcal{X}$; formula index

Definition. evaluation point

- u — deterministic point; $\mathcal{X}$; formula index

Definition. local-window point

- j — integer; $\{1, \dots, d\}$; formula index

Definition. coordinate index

- a — integer; $\{0, 1\}$; formula index

Definition. treatment-arm index

- t — scalar; $(0, \infty)$; formula index

Definition. margin argument

- q — scalar; $(0, \infty)$; approximation index

Definition. approximation smoothness order

- g — function; $\mathcal{X} \rightarrow \mathbb{R}$; approximation argument

Definition. generic target function

- v — vector; $\bigcup_{r \in \mathbb{N}} \mathbb{R}^r$; approximation coefficient

Definition. generic sieve coefficient

- X — random vector; $\mathcal{X}$; effect modifiers

Definition. covariates

- A — random variable; $\{0, 1\}$; treatment

Definition. treatment

- Y — random variable; $[-M_Y, M_Y]$; outcome

Definition. observed outcome

- Y^a — random variable; $[-M_Y, M_Y]$; potential outcome
Definition. potential outcome under arm a
- U_A — random variable; $[0,1]$; witness input
Definition. treatment uniform variate
- U_0 — random variable; $[0,1]$; witness input
Definition. control uniform variate
- U_1 — random variable; $[0,1]$; witness input
Definition. treated uniform variate
- Z_i — random vector; $\mathcal{X} \times \{0,1\} \times [-M_Y, M_Y]$; sample element
Definition. the i -th observed triple
- X_i — random vector; $\mathcal{X}$; sample component
Definition. covariate component of Z_i
- P — probability law; $\mathcal{P}(\mathcal{X} \times \{0,1\} \times [-M_Y, M_Y])$; statistical law
Definition. observed-data law
- P_X — probability law; $\mathcal{P}(\mathcal{X})$; covariate law
Definition. covariate marginal
- f_P — function; $\mathcal{X} \rightarrow \mathbb{R}_+$; covariate density
Definition. Lebesgue density of P_X
- F_0 — function; $\mathcal{X} \rightarrow \mathbb{R}_+$; estimator density input
Definition. known uniform density
- π — function; $\mathcal{X} \rightarrow [0,1]$; treatment mechanism
Definition. propensity score
- μ_0 — function; $\mathcal{X} \rightarrow \mathbb{R}$; nuisance regression
Definition. control outcome regression
- μ_1 — function; $\mathcal{X} \rightarrow \mathbb{R}$; outcome regression
Definition. treated outcome regression
- τ_P — function; $\mathcal{X} \rightarrow \mathbb{R}$; identified CATE
Definition. designated continuous CATE version
- $\mathcal{D}_P(\eta)$ — set; $2^{\mathcal{X}}$; local region
Definition. near-threshold band
- G_P — set; $2^{\mathcal{X}}$; causal target
Definition. CATE upper-level set
- H — set; $2^{\mathcal{X}}$; loss argument
Definition. generic measurable set

- d_H — loss function; $2^{\{\mathcal{X}\} \times 2^{\{\mathcal{X}\} \times \mathcal{P}(\mathbb{R}_+)}$; risk loss

Definition. weighted symmetric-difference loss

- $\widehat{G}_n$ — statistic; $2^{\{\mathcal{X}\}}$; decision estimator

Definition. measurable set estimator

- r_n^* — scalar; $(0, \infty)$; benchmark rate

Definition. two-branch CATE scale

- r_n — scalar; $(0, \infty)$; tuning sequence

Definition. tuning-rate alias

- $\mathcal{M}(\xi, \gamma; \kappa)$ — class of laws; $\mathcal{P}(\mathcal{X} \times \{0, 1\} \times [-M_Y, M_Y])$; minimax class

Definition. smooth CATE law class

- $\mathcal{M}_{\{\mathrm{tr}\}}(\gamma; \kappa)$ — class of laws; $\mathcal{P}(\mathcal{X} \times \{0, 1\} \times [-M_Y, M_Y])$; headline model

Definition. calibrated transverse class

- $\mathfrak{R}_n(\xi, \gamma; \kappa)$ — scalar; $\mathbb{R}_+$; minimax criterion

Definition. minimax weighted set risk

- $\mathcal{F}(\kappa)$ — set; $2^{\{(0, \infty)^2\}}$; open-problem target

Definition. feasible super-margin frontier

- $\mathcal{P}_{\{\mathrm{tr}\}}$ — augmented probability law; $\mathcal{P}(\mathcal{X} \times [0, 1]^3 \times \{0, 1\}^4)$; explicit witness

Definition. calibrated transverse causal witness

- p_a — function; $\mathcal{X} \rightarrow [0, 1]$; witness conditional mean

Definition. witness arm-specific Bernoulli probability

- h — scalar; $(0, 1]$; estimator tuning index

Definition. local bandwidth

- L_j — scalar; $[-1, 1]$; window geometry

Definition. left endpoint of coordinate window

- R_j — scalar; $[-1, 1]$; window geometry

Definition. right endpoint of coordinate window

- ℓ_j — scalar; $(0, 2]$; window geometry

Definition. coordinate-window length

- $W_{\{x, h\}}$ — set; $2^{\{\mathcal{X}\}}$; local neighborhood

Definition. truncated cube window

- $V_{\{x, h\}}$ — scalar; $(0, \infty)$; kernel normalization

Definition. window volume

- $T_{\{x,h\}}$ — map; $\mathcal{X} \rightarrow [0,1]^d$; boundary transformation
Definition. window rescaling map
- $K_{\{x,h\}}$ — function; $\mathcal{X} \rightarrow \mathbb{R}_+$; local weight
Definition. normalized cube kernel
- p — integer; $\mathbb{N}$; estimator tuning index
Definition. local polynomial order
- J_p — integer; $\mathbb{N}$; local basis dimension
Definition. number of total-degree polynomial terms
- ρ_p — vector-valued function; $[0,1]^d \rightarrow \mathbb{R}^{J_p}$; local polynomial basis
Definition. shifted-Legendre total-degree basis
- $r_{\{x,h,p\}}$ — vector-valued function; $\mathcal{X} \rightarrow \mathbb{R}^{J_p}$; local polynomial basis
Definition. rescaled local polynomial basis
- r_x — vector; $\mathbb{R}^{J_p}$; evaluation vector
Definition. local basis evaluated at x
- r_B — integer; $\mathbb{N}$; sieve order
Definition. B-spline order
- m — integer; $\mathbb{N}$; sieve tuning index
Definition. tensor-grid resolution
- k — integer; $\mathbb{N}$; sieve tuning index
Definition. sieve dimension
- B_k — vector-valued function; $[0,1]^d \rightarrow \mathbb{R}^k$; sieve basis
Definition. orthonormal tensor-product B-spline basis
- $b_{\{x,h,k\}}$ — vector-valued function; $\mathcal{X} \rightarrow \mathbb{R}^k$; higher-order basis
Definition. localized normalized spline basis
- $Q_{\{x,h\}}$ — matrix; $\mathbb{R}^{J_p \times J_p}$; local design matrix
Definition. population local Gram matrix
- $\Omega_{\{x,h,k\}}$ — matrix; $\mathbb{R}^{k \times k}$; higher-order design matrix
Definition. population localized spline Gram matrix
- K_0 — integer; $\{2,3,\dots\}$; cross-fitting index
Definition. number of cross-fitting folds
- ℓ — integer; $\{1,\dots,K_0\}$; cross-fitting index
Definition. fold index
- $\mathcal{I}_{\ell}$ — set; $2^{\{1,\dots,n\}}$; cross-fitting partition
Definition. evaluation-fold index set
- $\widehat{\pi}^{(-\ell)}$ — function; $\mathcal{X} \rightarrow [0,1]$; nuisance estimate
Definition. training-fold propensity estimate

- $\widehat{\mu}_0(-\ell)$ — function; $\mathcal{X} \rightarrow \mathbb{R}$; nuisance estimate
Definition. training-fold control-regression estimate
- $\widetilde{\Omega}^{(-\ell)}_{\{x,h,k\}}$ — matrix; $\mathbb{R}^{k \times k}$; matrix estimate
Definition. raw training spline Gram matrix
- $\widehat{\Omega}^{(-\ell)}_{\{x,h,k\}}$ — matrix; $\mathbb{R}^{k \times k}$; matrix estimate
Definition. spectrally stabilized spline Gram matrix
- q_- — scalar; $(0, \infty)$; matrix constant
Definition. local spectral lower threshold
- q_+ — scalar; $(0, \infty)$; matrix constant
Definition. local spectral upper threshold
- $\mathcal{Z}$ — set; $\mathcal{X} \times \{0,1\} \times [-M,Y,M,Y]$; operator domain
Definition. observed-data sample space
- q_1 — matrix-valued function; $\mathcal{Z} \rightarrow \mathbb{R}^{J_p \times J_p}$; estimating score
Definition. first-order local Gram score
- q_2 — matrix-valued kernel; $\mathcal{Z}^2 \rightarrow \mathbb{R}^{J_p \times J_p}$; estimating kernel
Definition. second-order local Gram score
- s_1 — vector-valued function; $\mathcal{Z} \rightarrow \mathbb{R}^{J_p}$; estimating score
Definition. first-order local response score
- s_2 — vector-valued kernel; $\mathcal{Z}^2 \rightarrow \mathbb{R}^{J_p}$; estimating kernel
Definition. second-order local response score
- $P_{-\ell}$ — empirical operator; $\{f: \mathcal{Z} \rightarrow \mathbb{R}\} \rightarrow \mathbb{R}$; empirical operator
Definition. evaluation-fold empirical average
- $U_{-\ell}$ — U-statistic operator; $\{f: \mathcal{Z}^2 \rightarrow \mathbb{R}\} \rightarrow \mathbb{R}$; empirical operator
Definition. ordered evaluation-fold second-order average
- $\widetilde{Q}^{(-\ell)}_{\{x,h\}}$ — matrix; $\mathbb{R}^{J_p \times J_p}$; matrix estimate
Definition. raw local Gram estimate
- $\widetilde{R}^{(-\ell)}_{\{x,h\}}$ — vector; $\mathbb{R}^{J_p}$; response estimate
Definition. raw local response estimate
- $\widehat{Q}^{(-\ell)}_{\{x,h\}}$ — matrix; $\mathbb{R}^{J_p \times J_p}$; matrix estimate
Definition. spectrally stabilized local Gram estimate
- $\widehat{\tau}_n$ — statistic; $\{g: \mathcal{X} \rightarrow \mathbb{R}\}$; CATE estimate
Definition. cross-fitted higher-order local-polynomial R-learner
- $\widehat{G}_n(\mathrm{LP})$ — statistic; $2^{\mathcal{X}}$; achieving estimator
Definition. thresholded boundary-adapted R-learner set estimator
- T — scalar; $(0, \infty)$; tuning index
Definition. low-smoothness tuning exponent

- $h.n$ — scalar; $(0,1]$; tuning sequence

Definition. chosen local bandwidth

- $C_{\{\mathrm{tun}\}}$ — scalar; $(0,\infty)$; tuning constant

Definition. fixed lower tuning constant

- $m.n$ — integer; $\mathbb{N}$; tuning sequence

Definition. chosen tensor-grid resolution

- $k.n$ — integer; $\mathbb{N}$; tuning sequence

Definition. chosen sieve dimension m_n^d

- $c_{.1}$ — scalar; $(0,\infty)$; theorem constant

Definition. lower comparison constant

- $C_{.1}$ — scalar; $(0,\infty)$; theorem constant

Definition. upper comparison constant

- $C_{\{\mathrm{app}\}}$ — scalar; $(0,\infty)$; approximation constant

Definition. uniform approximation constant

4 Assumptions

Assumption 1 (ass:iid-sampling). $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} P$.

Standard (*independent and identically distributed sampling*, [1]).

Assumption 2 (ass:consistency). $Y = Y^A$.

Standard (*consistency*, [1]).

Assumption 3 (ass:conditional-exchangeability). $(Y^0, Y^1) \perp A \mid X$.

Standard (*conditional exchangeability*, [1]).

Assumption 4 (ass:strict-overlap). $c_\pi \leq \pi(x) \leq 1 - c_\pi$ for $x \in \mathcal{X}$.

Standard (*strict overlap*, [1]).

Assumption 5 (ass:bounded-density).

$P_X(\mathcal{X}) = 1$ and $c_f \leq f_P(x) \leq C_f$ for Lebesgue-almost every $x \in \mathcal{X}$.

Standard (*two-sided bounded Lebesgue density on a fixed support*, [1]).

Assumption 6 (ass:continuous-cate-version). $\tau_P \in C(\mathcal{X})$.

Novel. *Continuous conditional-mean versions are natural on compact covariate supports; the calibrated witness has a linear CATE version.*

Assumption 7 (ass:near-band-holder). $\tau_P \in H_\gamma(L_\tau; \mathcal{D}_P(\eta))$.

Standard (*local Hölder smoothness*, [1]).

Assumption 8 (ass:off-band-holder). $\tau_P \in H_{\gamma'}(L_\tau; \mathcal{X} \setminus \mathcal{D}_P(\eta))$.

Standard (*off-band Hölder smoothness*, [1]).

Assumption 9 (ass:propensity-holder). $\pi \in H_\alpha(L_\pi; \mathcal{X})$.

Standard (*Hölder propensity model*, [2]).

Assumption 10 (ass:control-holder). $\mu_0 \in H_\beta(L_0; \mathcal{X})$.

Standard (*Hölder outcome-regression model*, [2]).

Assumption 11 (ass:lower-margin). $c_m t^\xi \leq P_X\{0 < |\tau_P(X) - \theta| \leq t\}$ for $0 < t \leq t_0$.

Novel. A two-sided lower margin mass records a nondegenerate boundary; in the calibrated transverse model the exponent is one.

Assumption 12 (ass:upper-margin). $P_X\{0 < |\tau_P(X) - \theta| \leq t\} \leq C_m t^\xi$ for $0 < t \leq t_0$.

Standard (Tsybakov upper-margin condition, [3]).

Assumption 13 (ass:interior-crossing). $\exists x^-, x^+ \in \text{int}(\mathcal{X}) : \tau_P(x^-) < \theta < \tau_P(x^+)$.

Standard (interior threshold crossing, [3]).

5 Definitions

Definition 1 (def:holder-ball). Defined object. $\backslash\mathit{mathsf{H}}\backslash\mathit{rho}(L;U)$

Construction. $\mathbf{H}_\rho(L;U) = \{g : U \rightarrow \mathbb{R} : \|g\|_{\mathbf{H}^\rho(U)} \leq L\}$.

Definition 2 (def:calibration). Defined object. *calibrated parameter value*

Construction.

$\mathcal{X} = [-1, 1]^d$, $M_Y = 1$, $\theta = 0$, $\eta = t_0 = c$, $c_f = C_f = 2^{-d}$, $c_\pi = 1/4$, $c_m = C_m = c^{-1}$, $L_\pi = L_0 = L_\tau = 1$.

Definition 3 (def:average-smoothness). Defined object. *average nuisance smoothness*

Construction. $\bar{s} = (\alpha + \beta)/2$.

Definition 4 (def:cate). Defined object. *designated CATE version*

Construction. $\tau_P(x) = \mu_1(x) - \mu_0(x)$ for P_X -almost every $x \in \mathcal{X}$, with τ_P the continuous representative.

Definition 5 (def:band). Defined object. *near-threshold band*

Construction. $\mathcal{D}_P(\eta) = \{x \in \mathcal{X} : |\tau_P(x) - \theta| \leq \eta\}$.

Definition 6 (def:upper-set). Defined object. *CATE upper-level set*

Construction. $G_P = \{x \in \mathcal{X} : \tau_P(x) > \theta\}$.

Definition 7 (def:weighted-loss). Defined object. *weighted symmetric-difference loss*

Construction. $d_H(H, G_P; P) = \int_{H \Delta G_P} |\tau_P(x) - \theta| f_P(x) dx$.

Definition 8 (def:rate). Defined object. *two-branch CATE scale*

Construction. $r_n^* = n^{-\gamma/(2\gamma+d)}$ if $\bar{s} \geq d/[4\{1 + d/(2\gamma)\}]$, and $r_n^* = n^{-1/[1+d/(4\bar{s})+d/(2\gamma)]}$ otherwise.

Definition 9 (def:smooth-model). Defined object. $\backslash\mathit{mathcal{M}}(\backslash\mathit{xi}, \backslash\mathit{gamma}; \backslash\mathit{kappa})$

Construction.

$\mathcal{M}(\xi, \gamma; \kappa) = \{P \in \mathcal{P}(\mathcal{X} \times \{0, 1\}) \times [-M_Y, M_Y] : P \text{ satisfies } \text{ass} : \text{strict-overlap}, \text{ass} : \text{bounded-density}, \text{ass} : \text{continuous-cate-version}, \text{ass} : \text{near-band-holder}, \text{ass} : \text{off-band-holder}, \text{ass} : \text{propensity-holder}, \text{ass} : \text{control-holder}, \text{ass} : \text{lower-margin}, \text{ass} : \text{upper-margin}, \text{ass} : \text{interior-crossing}\}$

Member properties. *ass:strict-overlap; ass:bounded-density; ass:continuous-cate-version; ass:near-band-holder; ass:off-band-holder; ass:propensity-holder; ass:control-holder; ass:lower-margin; ass:upper-margin; ass:interior-crossing.*

Definition 10 (def:transverse-model). Defined object. $\backslash\mathit{mathcal{M}}\backslash\mathit{tr}\backslash\mathit{gamma}; \backslash\mathit{kappa})$

Construction. $\mathcal{M}_{\text{tr}}(\gamma; \kappa) = \mathcal{M}(1, \gamma; \kappa)$ for $\gamma \in (d, \infty)$.

Member properties. *ass:strict-overlap; ass:bounded-density; ass:continuous-cate-version; ass:near-band-holder; ass:off-band-holder; ass:propensity-holder; ass:control-holder; ass:lower-margin; ass:upper-margin; ass:interior-crossing.*

Definition 11 (def:minimax-risk). Defined object. $\mathfrak{R}_n(\xi, \gamma; \kappa)$

Construction.

$$\mathfrak{R}_n(\xi, \gamma; \kappa) = \inf_{\widehat{G}_n} \sup_{P \in \mathcal{M}(\xi, \gamma; \kappa)} \mathbb{E}_P[d_H(\widehat{G}_n, G_P; P)]$$

Definition 12 (def:feasible-frontier). Defined object. $\mathcal{F}(\kappa)$

Construction.

$$\mathcal{F}(\kappa) = \{(\xi, \gamma) \in (0, \infty)^2 : \xi\gamma > d, \mathcal{M}(\xi, \gamma; \kappa) \neq \emptyset\}$$

Definition 13 (def:transverse-witness). Defined object. P^{tr}

Construction. Given $X = x$, draw U_A, U_0, U_1 independently as $\text{Unif}(0, 1)$, independently of X , and set $A = \mathbf{1}\{U_A \leq 1/2\}$, $Y^0 = \mathbf{1}\{U_0 \leq 1/2\}$, $Y^1 = \mathbf{1}\{U_1 \leq 1/2 + cx_1\}$, and $Y = Y^A$, with $X \sim \text{Unif}([-1, 1]^d)$. Equivalently, $P(A = a, Y^0 = y_0, Y^1 = y_1 \mid X = x) = \frac{1}{2} \prod_{b=0}^1 p_b(x)^{y_b} \{1 - p_b(x)\}^{1-y_b}$, where $p_0(x) = 1/2$ and $p_1(x) = 1/2 + cx_1$.

Definition 14 (def>window-basis). Defined object. *boundary-adapted local basis*

Construction.

$L_j = \max\{-1, x_j - h/2\}$, $R_j = \min\{1, x_j + h/2\}$, $\ell_j = R_j - L_j$, $W_{x,h} = \prod_{j=1}^d [L_j, R_j]$, $V_{x,h} = \prod_{j=1}^d \ell_j$, $T_{x,h}(u)_j = (u_j - L_j)/\ell_j$, $K_{x,h}(u) = V_{x,h}^{-1} \mathbf{1}\{u \in W_{x,h}\}$. Also $J_p = \binom{d+p}{p}$, $r_{x,h,p}(u) = \rho_p(T_{x,h}(u))$, and $r_x = r_{x,h,p}(x)$, where ρ_p is the shifted-Legendre total-degree basis orthonormal on $[0, 1]^d$.

Definition 15 (def:spline-basis). Defined object. *localized tensor B-spline basis*

Construction. Choose $r_B \geq \lceil \max\{\alpha, \beta, \gamma, \gamma'\} \rceil + 1$, let $k = m^d$, let B_k be an $L_2([0, 1]^d)$ -orthonormalized tensor-product B-spline basis of order r_B , and set $b_{x,h,k}(u) = V_{x,h}^{-1/2} B_k(T_{x,h}(u)) \mathbf{1}\{u \in W_{x,h}\}$.

Definition 16 (def:local-matrices). Defined object. *population local matrices*

Construction. $Q_{x,h} = \mathbb{E}[K_{x,h}(X) r_{x,h,p}(X) r_{x,h,p}(X)^\top \pi(X) \{1 - \pi(X)\}]$ and $\Omega_{x,h,k} = \mathbb{E}[b_{x,h,k}(X) b_{x,h,k}(X)^\top]$.

Definition 17 (def:boundary-adapted-estimator). Defined object. $\widehat{G}_n \sim \{\text{LP}\}$

Construction. Use the known density input $F_0(x) = 2^{-d}$, set $p = \lceil \gamma \rceil - 1$, $q_- = 2^{-d-1} c_\pi(1 - c_\pi)$, and $q_+ = 2^{-d-1}$, and partition $\{1, \dots, n\}$ into fixed balanced folds $\mathcal{I}_1, \dots, \mathcal{I}_{K_0}$. On every training complement set $\widehat{\pi}^{(-\ell)}(x) = 1/2$, $\widehat{\mu}_0^{(-\ell)}(x) = 0$, and $\widehat{\Omega}_{x,h,k}^{(-\ell)} = 2^{-d} I_k$. Let $P_\ell f = |\mathcal{I}_\ell|^{-1} \sum_{i \in \mathcal{I}_\ell} f(Z_i)$ and $U_\ell f = |\mathcal{I}_\ell|^{-1} (|\mathcal{I}_\ell| - 1)^{-1} \sum_{i \neq j \in \mathcal{I}_\ell} f(Z_i, Z_j)$. Define $q_1(Z) = K_{x,h}(X) r_{x,h,p}(X) r_{x,h,p}(X)^\top A [A - \widehat{\pi}^{(-\ell)}(X)]$,

$$q_2(Z_1, Z_2) = -V_{x,h} K_{x,h}(X_1) K_{x,h}(X_2) r_{x,h,p}(X_1) r_{x,h,p}(X_1)^\top [A_1 - \widehat{\pi}^{(-\ell)}(X_1)] b_{x,h,k}(X_1)^\top [\widehat{\Omega}_{x,h,k}^{(-\ell)}]^{-1} b_{x,h,k}(X_2) A_2$$

, $s_1(Z) = K_{x,h}(X) r_{x,h,p}(X) [Y - \widehat{\mu}_0^{(-\ell)}(X)] [A - \widehat{\pi}^{(-\ell)}(X)]$, and

$$s_2(Z_1, Z_2) = -V_{x,h} K_{x,h}(X_1) K_{x,h}(X_2) r_{x,h,p}(X_1) [A_1 - \widehat{\pi}^{(-\ell)}(X_1)] b_{x,h,k}(X_1)^\top [\widehat{\Omega}_{x,h,k}^{(-\ell)}]^{-1} b_{x,h,k}(X_2) [Y_2 - \widehat{\mu}_0^{(-\ell)}(X_2)]$$

. Set $\widetilde{Q}_{x,h}^{(\ell)} = P_\ell q_1 + U_\ell q_2$, $\widetilde{R}_{x,h}^{(\ell)} = P_\ell s_1 + U_\ell s_2$, and

$$\widehat{Q}_{x,h}^{(\ell)} = T_{[q_-, q_+]} \{(\widetilde{Q}_{x,h}^{(\ell)} + \widetilde{Q}_{x,h}^{(\ell)\top})/2\}$$

. For symmetric $M = V \text{diag}(\lambda_j) V^\top$, let $T_{[a,b]}(M) = V \text{diag}(\min\{b, \max\{a, \lambda_j\}\}) V^\top$. Finally set

$$\widehat{\tau}_n(x) = K_0^{-1} \sum_{\ell=1}^{K_0} \text{clip}_{[-2M_Y, 2M_Y]} \{r_x^\top [\widehat{Q}_{x,h}^{(\ell)}]^{-1} \widetilde{R}_{x,h}^{(\ell)}\}$$

and $\widehat{G}_n^{\text{LP}} = \{x \in \mathcal{X} : \widehat{\tau}_n(x) > \theta\}$. A nonfinite raw matrix or a fold of size smaller than two returns θ on that fold.

Inputs. Z_i ; κ .

Definition 18 (def:tuning). Defined object. *two-branch tuning*

Construction. Let $\bar{s} = (\alpha + \beta)/2$, $T = 1 + d/(4\bar{s}) + d/(2\gamma)$, and $r_n = r_n^*$. In the high-smoothness branch set $h_n = n^{-1/(2\gamma+d)}$, $m_n = \lceil C_{\text{tun}}^{1/d} (nh_n^d)^{1/d} \rceil$, and $k_n = m_n^d$. In the low-smoothness branch set $h_n = n^{-1/(\gamma T)}$, $m_n = \lceil C_{\text{tun}}^{1/d} h_n (\tau_n^*)^{-1/(2\bar{s})} \rceil$, and $k_n = m_n^d$.

6 Main results

Theorem 1 (thm:identified-version). Under $\text{ass} : \text{consistency}$, $\text{ass} : \text{conditional} - \text{exchangeability}$, $\text{ass} : \text{strict} - \text{overlap}$, and $\text{ass} : \text{continuous} - \text{cate} - \text{version}$, $\tau_P(X) = \mathbb{E}[Y^1 - Y^0 \mid X]$ almost surely. The continuous representative is unique on $\text{supp}(P_X)$; hence G_P is an observed-law functional modulo P_X -null sets, and $d_H(H, G_P; P)$ is invariant to P_X -null changes of either set.

Proof. Identification. Fix $a \in \{0, 1\}$. Boundedness of Y^a guarantees integrability. By $\text{ass} : \text{conditional} - \text{exchangeability}$,

$$\mathbb{E}_P[Y^a \mid X, A = a] = \mathbb{E}_P[Y^a \mid X]$$

where the left-hand conditional mean is defined. By $\text{ass} : \text{strict} - \text{overlap}$, $P(A = a \mid X) \geq c_\pi > 0$ almost surely, so the arm-specific conditional mean is determined for P_X -almost every covariate value. On $\{A = a\}$, $\text{ass} : \text{consistency}$ gives $Y = Y^a$. Therefore

$$\mu_a(X) = \mathbb{E}_P[Y \mid X, A = a] = \mathbb{E}_P[Y^a \mid X, A = a] = \mathbb{E}_P[Y^a \mid X] \quad P\text{-almost surely.}$$

Subtracting the identities for $a = 0$ and $a = 1$, and then applying $\text{def} : \text{cate}$, yields

$$\tau_P(X) = \mu_1(X) - \mu_0(X) = \mathbb{E}_P[Y^1 - Y^0 \mid X] \quad P\text{-almost surely.}$$

Uniqueness on the support. Let $g_1, g_2 \in C(\mathcal{X})$ represent the same P_X -almost-sure equivalence class. Suppose that $g_1(x_0) \neq g_2(x_0)$ at some $x_0 \in \text{supp}(P_X)$. Continuity gives an $\varepsilon > 0$ and a relatively open neighborhood O of x_0 in $\mathcal{X}$ such that

$$|g_1(x) - g_2(x)| > \varepsilon \quad \text{for every } x \in O.$$

The definition of support gives $P_X(O) > 0$, contradicting $g_1 = g_2$ P_X -almost surely. Hence $g_1 = g_2$ on $\text{supp}(P_X)$. By $\text{ass} : \text{continuous} - \text{cate} - \text{version}$, this applies to the designated continuous representative τ_P .

The observed law determines each μ_a as an arm-specific conditional mean on the set where that arm has positive conditional probability. Strict overlap makes this set have full P_X -measure for both arms. Thus the observed law determines τ_P P_X -almost surely, and the preceding uniqueness argument determines its continuous representative on $\text{supp}(P_X)$. By $\text{def} : \text{upper} - \text{set}$, any two versions of G_P obtained from observationally equivalent representatives can therefore differ only on a P_X -null set.

Null-set invariance of the loss. Because f_P is the Lebesgue density of P_X , $\text{def} : \text{weighted} - \text{loss}$ can be written as

$$d_H(H, G_P; P) = \int_{\mathcal{X}} \mathbf{1}_{H \Delta G_P}(x) |\tau_P(x) - \theta| dP_X(x).$$

Let H' and G'_P satisfy $P_X(H' \Delta H) = 0$ and $P_X(G'_P \Delta G_P) = 0$. The elementary set inclusion

$$(H' \Delta G'_P) \Delta (H \Delta G_P) \subseteq (H' \Delta H) \cup (G'_P \Delta G_P)$$

shows that the two indicator functions in the corresponding loss integrals agree P_X -almost surely. Their nonnegative integrals are consequently equal. This proves invariance to P_X -null changes of either set. No numerical upper or lower bound on f_P is used. $\square$

Justification. This theorem separates the causal CATE target from its observed-law conditional-mean representation. *Gap.* Identification is needed before the level set and weighted loss can be treated as observed-law objects. *Consumer.* It supports the causal interpretation of every surviving feasibility and risk statement.

Theorem 2 (thm:calibrated-witness). Under $\text{def} : \text{calibration}$, the observed marginal induced by P^{tr} belongs to $\mathcal{M}_{\text{tr}}(\gamma; \kappa)$ for every $\gamma > d$ and $\gamma' > 0$. It satisfies $(Y^0, Y^1) \perp A \mid X$, $Y = Y^A$, $\pi = 1/2$, $\mu_0 = 1/2$, $\mu_1(x) = 1/2 + cx_1$, $\tau_{P^{\text{tr}}}(x) = cx_1$, and $P_X\{0 < |\tau_{P^{\text{tr}}}(X) - \theta| \leq t\} = t/c$ for $0 < t \leq c$.

Proof. By def : transverse – witness, conditional on $X = x$, the treatment variable A is a function of U_A , whereas (Y^0, Y^1) is a function of (U_0, U_1) . The three uniforms are conditionally independent, so

$$(Y^0, Y^1) \perp A \mid X.$$

The same definition sets $Y = Y^A$. Since $A = \mathbf{1}\{U_A \leq 1/2\}$,

$$\pi(x) = P(A = 1 \mid X = x) = \frac{1}{2}.$$

Conditional exchangeability and consistency, or directly the independent-uniform construction, give

$$\mu_0(x) = \mathbb{E}[Y \mid A = 0, X = x] = \frac{1}{2}, \quad \mu_1(x) = \mathbb{E}[Y \mid A = 1, X = x] = \frac{1}{2} + cx_1.$$

Consequently, def : cate gives the continuous representative

$$\tau_{P^{\text{tr}}}(x) = \mu_1(x) - \mu_0(x) = cx_1.$$

Because $0 < c < 1/4$ and $x_1 \in [-1, 1]$, all Bernoulli probabilities in the construction belong to $[1/4, 3/4]$, and $Y \in [0, 1] \subseteq [-M_Y, M_Y]$ under $M_Y = 1$. The propensity $1/2$ satisfies strict overlap with $c_\pi = 1/4$. The uniform distribution on $[-1, 1]^d$ has density $2^{-d} = c_f = C_f$.

The functions π and μ_0 are constant, while $\tau_{P^{\text{tr}}}(x) = cx_1$ is affine. In the Taylor-remainder norm of def : holder – ball, the relevant norms are at most one: the only nonzero derivative of the affine term has magnitude $c < 1/4$, and every higher-order Taylor remainder is zero. Furthermore,

$$|\tau_{P^{\text{tr}}}(x) - \theta| = |cx_1| \leq c = \eta,$$

so $\mathcal{D}_{P^{\text{tr}}}(\eta) = \mathcal{X}$. Thus the near-band Hölder condition holds for every $\gamma > d$, and the off-band Hölder condition is vacuous for every $\gamma' > 0$.

For $0 < t \leq c$, uniformity of X_1 and $\theta = 0$ yield

$$\begin{aligned} P_X\{0 < |\tau_{P^{\text{tr}}}(X) - \theta| \leq t\} &= P\{0 < |cX_1| \leq t\} \\ &= P\{0 < |X_1| \leq t/c\} = \frac{t}{c}. \end{aligned}$$

Since $c_m = C_m = c^{-1}$, this identity is exactly the lower and upper margin condition with $\xi = 1$ for $0 < t \leq t_0 = c$. Finally, the interior points

$$x^- = (-1/2, 0, \dots, 0), \quad x^+ = (1/2, 0, \dots, 0)$$

have CATE values $-c/2 < 0 = \theta < c/2$. Hence the interior-crossing condition holds. We have verified every member property in def : smooth – model, so the observed marginal induced by P^{tr} belongs to

$$\mathcal{M}(1, \gamma; \kappa) = \mathcal{M}_{\text{tr}}(\gamma; \kappa)$$

for every $\gamma > d$ and every $\gamma' > 0$, and all the asserted identities have already been established. $\square$

Justification. The explicit transverse law proves nonemptiness and an interior threshold crossing for every feasible calibrated smoothness index. *Gap.* A feasibility characterization requires an actual member law, not only a compatibility obstruction. *Consumer.* It supplies the reverse inclusion in the calibrated frontier theorem and the nonemptiness part of risk well-posedness.

Lemma 1 (lem:uniform-design). *For every $P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)$ under def : calibration, $f_P = F_0 = 2^{-d}$ Lebesgue-almost everywhere on $\mathcal{X}$; its zero extension vanishes off $\mathcal{X}$. Thus the density-ratio error is zero and the density-ratio supremum is one.*

Proof. Fix $P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)$. Under def : calibration, the support is $\mathcal{X} = [-1, 1]^d$ and the two density constants satisfy $c_f = C_f = 2^{-d}$. Assumption ass : bounded – density therefore gives

$$2^{-d} \leq f_P(x) \leq 2^{-d}$$

for Lebesgue-almost every $x \in \mathcal{X}$, whence $f_P(x) = 2^{-d}$ almost everywhere on $\mathcal{X}$. By its declared meaning, F_0 is the uniform density on $\mathcal{X}$; since the cube has volume 2^d , normalization gives $F_0(x) = 2^{-d}$. Thus $f_P = F_0$ almost everywhere. By definition, the zero extension of a density supported on $\mathcal{X}$ vanishes on $\mathbb{R}^d \setminus \mathcal{X}$. Consequently $f_P/F_0 = 1$ almost everywhere on $\mathcal{X}$: any density-ratio error relative to F_0 is zero and the essential supremum of the ratio is one. $\square$

Justification. Calibration and the two-sided density equality force the covariate density to equal the known uniform reference. *Gap.* The displayed statistic uses a known density input whose relation to member laws must be explicit. *Consumer.* It supports the deterministic Gram and score-envelope calculations without implying a risk rate.

Open-ended Question 1 (oeq:nontransverse-frontier). *For fixed calibrated κ and $\gamma > d$, does $\mathfrak{R}_n(1, \gamma; \kappa) \asymp (r_n^*)^2$ hold in either or both branches of r_n^* ?*

For every fixed calibrated κ , every $\gamma > d$, and all sufficiently large n ,

$$cn^{-2\gamma/(2\gamma+d)} \leq \mathfrak{R}_n(1, \gamma; \kappa) \leq Cn^{-2 \min\{\beta, \gamma\}/\{2 \min\{\beta, \gamma\}+d\}}.$$

If $\beta \geq \gamma$, then $\bar{s} = (\alpha + \beta)/2 \geq \gamma/2 > d/[4\{1 + d/(2\gamma)\}]$, so the high-smoothness branch applies and $(r_n^*)^2 = n^{-2\gamma/(2\gamma+d)}$. Hence

$$\mathfrak{R}_n(1, \gamma; \kappa) \asymp (r_n^*)^2$$

in that subregime. When $\beta < \gamma$, the displayed two-sided bracket is the strongest conclusion established by the frozen proof spine.

Justification. This question now isolates the known-uniform-design minimax problem left after the matched $\beta \geq \gamma$ subregime. *Gap.* For $\beta < \gamma$, the direct bracket does not match, and the standard low-smoothness fuzzy construction violates the calibrated full-support density. *Consumer.* A future solution must neutralize the transition-region likelihood under uniform design or prove a different sharp rate for the calibrated class.

Lemma 2 (lem:exact-margin-graph-packing). *Under def : calibration, for every $\gamma > d$ and every sufficiently small $h > 0$, put $\delta = h^\gamma$. There are an integer $M \geq c_0 h^{-(d-1)}$, with $M = 1$ when $d = 1$, and functions $\tau_\omega \in \mathcal{H}_\gamma(1; \mathcal{X})$, indexed by $\omega \in \{0, 1\}^M$, such that: each τ_ω crosses zero in the interior; under the uniform law on $\mathcal{X}$,*

$$P_X\{0 < |\tau_\omega(X)| \leq t\} = t/c \quad (0 < t \leq c);$$

and, if $\omega^{(j)}$ differs from ω only in coordinate j , then the corresponding upper sets have a local weighted separation $s_j \geq c_1 \delta^2 h^{d-1}$ in the sense that, for every measurable H , the sum of the two losses restricted to the j -th patch is at least s_j . Moreover, adjacent CATEs differ by at most $C\delta$ on a set of Lebesgue measure at most Ch^d .

Proof. Put $D = d - 1$ and write $x = (x_1, z)$, where $z \in [-1, 1]^D$; when $d = 1$, all notation involving z is omitted. Choose a nonincreasing function $\chi \in C^\infty([0, \infty); [0, 1])$ such that $\chi = 1$ on $[0, 1/4]$ and $\chi = 0$ on $[3/4, \infty)$, and put $\psi = -2\chi\chi' \geq 0$. If $D \geq 1$, choose $\phi \in C_c^\infty((-1/2, 1/2)^D; [0, 1])$ equal to one on $[-1/4, 1/4]^D$. For every sufficiently small h , one can pack pairwise disjoint cubes $z_j + h(-1/2, 1/2)^D$ in $(-1, 1)^D$, $1 \leq j \leq M$, with $M \geq c_0 h^{-D}$. Fix a sufficiently small constant $a > 0$, independent of h and ω , and define

$$b_\omega(z) = a\delta \sum_{j=1}^M (2\omega_j - 1) \phi\left(\frac{z - z_j}{h}\right)^2, \quad \delta = h^\gamma.$$

For $d = 1$, take $M = 1$ and $b_\omega = a\delta(2\omega_1 - 1)$. Disjointness of the bump supports gives, for every multi-index q ,

$$\|D^q b_\omega\|_\infty \leq C_q a h^{\gamma - |q|}.$$

In particular, because $\gamma > d \geq 1$,

$$\frac{\|b_\omega\|_\infty}{h} \leq C a h^{\gamma-1} \longrightarrow 0.$$

This limit is the scale separation needed both for the change-of-variables construction and for the linear formulas used below.

Set $q_h(w) = \chi(|w|/h)^2$. This is smooth at zero because it is constant near zero. For fixed z , define

$$\Phi_\omega(w, z) = w + b_\omega(z)q_h(w), \quad -1 \leq w \leq 1.$$

For $w > 0$ and $w < 0$, respectively,

$$\partial_w \Phi_\omega(w, z) = 1 - \frac{b_\omega(z)}{h} \psi(w/h), \quad \partial_w \Phi_\omega(w, z) = 1 + \frac{b_\omega(z)}{h} \psi((-w)/h).$$

Thus, after decreasing the upper bound on h , $\partial_w \Phi_\omega \geq 1/2$ uniformly. Moreover, $\Phi_\omega(-1, z) = -1$ and $\Phi_\omega(1, z) = 1$. Hence $w \mapsto \Phi_\omega(w, z)$ is a smooth increasing bijection of $[-1, 1]$. Let $w_\omega(x_1, z)$ be its inverse and define

$$\tau_\omega(x_1, z) = c w_\omega(x_1, z).$$

The implicit-function theorem and the uniform lower bound on $\partial_w \Phi_\omega$ show that this is a smooth function on the cube. Its zero set is the graph $x_1 = b_\omega(z)$, and therefore

$$\{\tau_\omega > 0\} = \{(x_1, z) : x_1 > b_\omega(z)\}.$$

Because $q_h(w) = 0$ for $|w| \geq 3h/4$, one has $\tau_\omega(x) = cx_1$ away from an $O(h)$ -neighborhood of that graph. In particular, for small h , the two interior points $(1/2, 0)$ and $(-1/2, 0)$ have strictly positive and strictly negative values, respectively.

We next verify the exact margin identity. For $0 \leq v \leq 1$, the two points having $|w_\omega| = v$ are

$$x_+(v, z) = v + b_\omega(z)\chi(v/h)^2, \quad x_-(v, z) = -v + b_\omega(z)\chi(v/h)^2.$$

Their derivatives are

$$\partial_v x_+ = 1 - \frac{b_\omega(z)}{h} \psi(v/h), \quad \partial_v x_- = -1 - \frac{b_\omega(z)}{h} \psi(v/h).$$

Conditional on z , the calibrated uniform distribution gives density $1/2$ to X_1 . Therefore the conditional density of $V = |w_\omega(X_1, z)|$ is

$$f_{V|z}(v) = \frac{1}{2} (\partial_v x_+(v, z) + |\partial_v x_-(v, z)|) = 1, \quad 0 < v < 1.$$

Since $|\tau_\omega(X)| = cV$, integration over z yields

$$P_X\{0 < |\tau_\omega(X)| \leq t\} = \frac{t}{c}, \quad 0 < t \leq c.$$

It remains to check the claimed Hölder radius, including noninteger γ . Write $R_\omega = \tau_\omega - cx_1$. Repeated implicit differentiation of

$$x_1 = w_\omega + b_\omega(z)q_h(w_\omega)$$

uses only the bounds $\partial_w \Phi_\omega \geq 1/2$, $\|D^q b_\omega\|_\infty \leq C_q a h^{\gamma-|q|}$, and $\|q_h^{(r)}\|_\infty \leq C_r h^{-r}$. Induction on total derivative order consequently gives

$$\|D^m R_\omega\|_\infty \leq C_m a h^{\gamma-m} \quad \text{for every integer } 0 \leq m \leq \lceil \gamma \rceil.$$

For completeness, the induction step follows by differentiating the implicit equation once more: the unique highest-order derivative of w_ω is multiplied by $\partial_w \Phi_\omega$, while every remaining product has total scale $ah^{\gamma-m}$; division by $\partial_w \Phi_\omega \geq 1/2$ preserves that order.

If γ is an integer, the bound with $m = \gamma$ and Taylor's integral remainder give a degree- $\gamma - 1$ remainder bounded by $Ca\|x - y\|^\gamma$. If γ is not an integer, write $r = \lceil \gamma \rceil$ and $\nu = \gamma - r \in (0, 1)$. When $\|x - y\| \leq h$, the mean-value theorem and the $(r + 1)$ -st derivative bound give

$$\|D^r R_\omega(x) - D^r R_\omega(y)\| \leq C a h^{\nu-1} \|x - y\| \leq C a \|x - y\|^\nu.$$

When $\|x - y\| > h$, the r -th derivative bound instead gives

$$\|D^r R_\omega(x) - D^r R_\omega(y)\| \leq 2C a h^\nu \leq 2C a \|x - y\|^\nu.$$

This two-scale argument supplies the required order- γ Taylor remainder uniformly across bump boundaries. Under the Taylor-remainder norm in $\text{def} : \text{holder} - \text{ball}$, the linear map $x \mapsto cx_1$ has norm at most $2c < 1/2$. Taking a small enough, depending only on the fixed constants and γ, d , makes the uniform contribution of R_ω smaller than the remaining radius. Hence $\tau_\omega \in H_\gamma(1; \mathcal{X})$ for every ω .

Now fix an edge $(\omega, \omega^{(j)})$. On the core cube $z_j + h[-1/4, 1/4]^D$, the two boundary graphs are $b^+(z) = a\delta$ and $b^-(z) = -a\delta$, up to relabeling. Since $\delta/h = h^{\gamma-1} \rightarrow 0$, for all $-a\delta \leq x_1 \leq a\delta$ the inverse points satisfy $|w| \leq 2a\delta < h/4$, where $q_h = 1$. Thus on this core strip

$$\tau_+(x) = c(x_1 - a\delta), \quad \tau_-(x) = c(x_1 + a\delta).$$

At every point between the two graphs, an arbitrary measurable set H disagrees with at least one of the two upper sets. If the point belongs to H , its contribution to the loss for the upper graph is $c(a\delta - x_1)$; if it does not belong to H , its contribution to the loss for the lower graph is $c(x_1 + a\delta)$. Using the calibrated density 2^{-d} , the sum of the two losses restricted to this patch is at least

$$2^{-d} c \int_{z_j + h[-1/4, 1/4]^D} \int_{-a\delta}^{a\delta} \min\{x_1 + a\delta, a\delta - x_1\} dx_1 dz = c_1 \delta^2 h^D,$$

where $c_1 = 2^{-d} c a^2 2^{-D} > 0$; for $D = 0$, the tangential factor is one. This is the asserted lower bound $c_1 \delta^2 h^{d-1}$.

Finally, adjacent boundary functions differ only on one tangential cube of volume $O(h^{d-1})$. If b and b' are the adjacent graphs, monotonicity of Φ_b and $\partial_w \Phi_b \geq 1/2$ imply, at every common (x_1, z) ,

$$|w_b(x_1, z) - w_{b'}(x_1, z)| \leq 2|b(z) - b'(z)| \leq C\delta.$$

The inverses, and hence the CATEs, coincide unless z lies in that tangential cube and $|x_1| \leq Ch$, because $q_h = 0$ outside the normal transition strip and $\|b\|_\infty = o(h)$. Thus adjacent CATEs differ by at most $C\delta$ on a set of Lebesgue measure at most $Ch^{d-1}h = Ch^d$. All constants are independent of h, M , and ω , completing the proof. $\square$

Lemma 3 (lem:calibrated-margin-compatibility). *Under $\text{def} : \text{calibration}$, if $P \in \mathcal{M}(\xi, \gamma; \kappa)$ and $\xi\gamma > d$, then $\xi = 1$ and $\gamma > d$.*

Proof. Let $P \in \mathcal{M}(\xi, \gamma; \kappa)$ and suppose $\xi\gamma > d$. Membership in $\text{def} : \text{smooth} - \text{model}$, together with $c_m = C_m = c^{-1}$ and $t_0 = c$ from $\text{def} : \text{calibration}$, makes the lower and upper margin bounds coincide:

$$P_X\{0 < |\tau_P(X) - \theta| \leq t\} = c^{-1}t^\xi, \quad 0 < t \leq c. \quad (1)$$

At $t = c$, the left-hand side is at most one. Hence $c^{\xi-1} \leq 1$. Since $0 < c < 1$, this implies

$$\xi \geq 1. \quad (2)$$

We next obtain the reverse restriction from the interior crossing. Put $q = \min\{\gamma, 1\}$. Let x^- and x^+ be the interior points in $\text{ass} : \text{interior} - \text{crossing}$. The cube $\mathcal{X} = [-1, 1]^d$ is convex, and its interior contains the compact segment joining these two points. By continuity of τ_P , a sufficiently thin solid cylinder around a slightly shortened version of this segment can be parameterized as

$$\mathcal{C} = \{z + se : z \in B, 0 \leq s \leq \ell_0\} \subset \text{int}(\mathcal{X}),$$

where e is a unit vector, B is a $(d-1)$ -dimensional cross-section of positive measure, and there is $\varepsilon > 0$ such that, uniformly in $z \in B$,

$$\tau_P(z) - \theta \leq -\varepsilon, \quad \tau_P(z + \ell_0 e) - \theta \geq \varepsilon. \quad (3)$$

For $d = 1$, the cross-section consists of one point and its transverse measure is interpreted as one.

The zero set of $\tau_P - \theta$ is compact. Since $\eta = c > 0$, continuity supplies a fixed neighborhood of that zero set contained in $\mathcal{D}_P(\eta)$. On this neighborhood, $\text{ass} : \text{near} - \text{band} - \text{holder}$ implies

$$|\tau_P(x) - \tau_P(y)| \leq L_* \|x - y\|^q \quad (4)$$

whenever the segment joining x and y remains in the neighborhood, for a finite constant L_* . Indeed, when $\gamma \leq 1$, (4) is the defining Hölder bound; when $\gamma > 1$, the first derivatives are bounded by the Hölder norm and (4) follows with $q = 1$ from the mean-value theorem.

For each $z \in B$, let

$$s_z = \inf\{s \in [0, \ell_0] : \tau_P(z + se) - \theta = 0\}.$$

Equation (3) and continuity show that s_z exists and that $\tau_P(z + se) - \theta < 0$ for $0 \leq s < s_z$. Uniform continuity on the compact cylinder, together with the uniform endpoint separation in (3), gives constants $r_0 > 0$ and $t_1 > 0$, independent of z , such that the interval $[s_z - r_0, s_z]$ lies in the cylinder and in the neighborhood where (4) applies, and $(t/L_*)^{1/q} \leq r_0$ for $0 < t \leq t_1$. Therefore, for every such t ,

$$s_z - (t/L_*)^{1/q} \leq s < s_z \implies 0 < |\tau_P(z + se) - \theta| \leq t.$$

Fubini's theorem and the calibrated lower density bound $f_P \geq 2^{-d}$ now give a constant $C_* > 0$ such that

$$P_X\{0 < |\tau_P(X) - \theta| \leq t\} \geq C_* t^{1/q}, \quad 0 < t \leq t_1. \quad (5)$$

Combining (1) and (5) and letting $t \downarrow 0$ yields

$$\xi \leq \frac{1}{q} = \frac{1}{\min\{\gamma, 1\}}. \quad (6)$$

If $\gamma \leq 1$, then (6) gives $\xi\gamma \leq 1 \leq d$, contradicting $\xi\gamma > d$. Thus $\gamma > 1$, so (6) becomes $\xi \leq 1$. Together with (2), this proves $\xi = 1$. The assumed super-margin inequality then reduces to $\gamma > d$, as required. $\square$

Justification. Positive density, an interior crossing, local Hölder regularity, and the exact two-sided margin determine the only feasible calibrated exponent. *Gap.* Nominal parameter labels need not correspond to nonempty law classes. *Consumer.* It gives the forward inclusion in the exact calibrated frontier characterization.

Theorem 3 (thm:calibrated-risk-well-posedness). *For every fixed calibrated κ , every $\gamma > d$, and every n , the class $\mathcal{M}_{\text{tr}}(\gamma; \kappa)$ is nonempty and*

$$0 \leq \mathfrak{R}_n(1, \gamma; \kappa) \leq 2.$$

The upper inequality is witnessed by the constant empty-set estimator.

Proof. Fix $\gamma > d$ and $n \in \mathbb{N}$. By thm : calibrated – witness, $\mathcal{M}_{\text{tr}}(\gamma; \kappa)$ is nonempty. By def : transverse – model, it equals $\mathcal{M}(1, \gamma; \kappa)$, so the supremum in def : minimax – risk is over a nonempty class. Since the integrand in def : weighted – loss is nonnegative, every risk is nonnegative and hence

$$\mathfrak{R}_n(1, \gamma; \kappa) \geq 0.$$

For the upper bound, take the deterministic, sample-independent measurable estimator $\hat{G}_n = \emptyset$. Under calibration, $M_Y = 1$ and $\theta = 0$. Since $Y \in [-1, 1]$, each conditional mean $\mu_a(X)$ belongs to $[-1, 1]$ almost surely. Therefore def : cate gives $|\tau_P(X)| \leq 2$ almost surely for every $P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)$. Consequently,

$$\begin{aligned} d_H(\emptyset, G_P; P) &= \int_{G_P} |\tau_P(x) - \theta| f_P(x) dx \\ &\leq 2 \int_{\mathcal{X}} f_P(x) dx = 2. \end{aligned}$$

The estimator is deterministic, so taking its expectation does not change this bound. Taking the supremum over $P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)$ and then using this estimator as a candidate in the infimum of def : minimax – risk yields

$$\mathfrak{R}_n(1, \gamma; \kappa) \leq 2.$$

Both inequalities hold for every n , completing the proof. $\square$

Justification. The explicit transverse witness proves nonemptiness, and the constant empty-set estimator proves finite minimax risk for every sample size. *Gap.* The bound $0 \leq \mathfrak{R}_n(1, \gamma; \kappa) \leq 2$ alone does not imply consistency. *Consumer.* The rate results operate on a nonempty class with a finite decision problem.

Lemma 4 (lem:calibrated-direct-assouad-lower). *For every fixed calibrated κ , every $\gamma > d$, and all sufficiently large n ,*

$$\mathfrak{R}_n(1, \gamma; \kappa) \geq cn^{-2\gamma/(2\gamma+d)}$$

for a constant $c > 0$ independent of n .

Proof. Fix the calibrated κ and $\gamma > d$. For a bandwidth $h > 0$ to be selected below, put $\delta = h^\gamma$ and take the family $\{\tau_\omega : \omega \in \{0, 1\}^M\}$ supplied by lem : exact – margin – graph – packing. For each ω , define an observed-data law P_ω as follows. Let X be uniform on $[-1, 1]^d$, let $A \mid X \sim \text{Bernoulli}(1/2)$, and, conditionally on $(X = x, A = a)$, let $Y \in \{-1, 1\}$ satisfy

$$P_\omega(Y = 1 \mid X = x, A = 0) = \frac{1}{2}, \quad P_\omega(Y = 1 \mid X = x, A = 1) = \frac{1 + \tau_\omega(x)}{2}.$$

The exact margin identity at $t = c$ gives

$$P_X\{0 < |\tau_\omega(X)| \leq c\} = 1.$$

Since τ_ω is continuous and the uniform density is strictly positive on every relatively open subset of the cube, this implies $|\tau_\omega(x)| \leq c < 1/4$ for every x . Hence the displayed Bernoulli probabilities are valid and bounded away from zero and one.

We verify membership rather than building it into the construction. Under P_ω ,

$$\pi(x) = \frac{1}{2}, \quad \mu_0(x) = 0, \quad \mu_1(x) = \tau_\omega(x), \quad \tau_{P_\omega}(x) = \tau_\omega(x).$$

Thus strict overlap, the calibrated uniform density, the propensity Hölder condition, and the control-regression Hölder condition all hold. The preceding range bound gives $\mathcal{D}_{P_\omega}(c) = \mathcal{X}$, so the global membership $\tau_\omega \in \mathcal{H}_\gamma(1; \mathcal{X})$ from the packing lemma verifies the near-band condition, while the off-band condition is vacuous on the empty complement. The exact identity

$$P_X\{0 < |\tau_\omega(X)| \leq t\} = t/c, \quad 0 < t \leq c,$$

verifies both calibrated margin inequalities, and the packing lemma supplies the interior crossing. Outcomes lie in $[-1, 1]$, and the designated CATE is continuous. Therefore

$$P_\omega \in \mathcal{M}_{\text{tr}}(\gamma; \kappa) \quad \text{for every } \omega.$$

We next control adjacent testing affinity. If $\omega^{(j)}$ differs from ω only in coordinate j , the packing lemma gives

$$|\tau_\omega(x) - \tau_{\omega^{(j)}}(x)| \leq C\delta$$

on a set of Lebesgue measure at most Ch^d , and the two functions agree elsewhere. Let

$$\rho(P, Q) = \int \sqrt{dP dQ}$$

denote Hellinger affinity. For probabilities $p, q \in [3/8, 5/8]$,

$$(\sqrt{p} - \sqrt{q})^2 \leq \frac{(p - q)^2}{\min\{p, q\}} \leq C(p - q)^2,$$

and the same inequality applies to $1 - p$ and $1 - q$. Conditioning on (X, A) , noting that only the treated outcome law differs, and integrating over the uniform design therefore yield

$$1 - \rho(P_\omega, P_{\omega^{(j)}}) \leq C \int_{\mathcal{X}} \{\tau_\omega(x) - \tau_{\omega^{(j)}}(x)\}^2 dx \leq C\delta^2 h^d.$$

For the independent sample laws, affinity tensorizes:

$$\rho(P_\omega^{\otimes n}, P_{\omega^{(j)}}^{\otimes n}) = \rho(P_\omega, P_{\omega^{(j)}})^n,$$

so $1 - \rho(P_\omega^{\otimes n}, P_{\omega^{(j)}}^{\otimes n}) \leq Cn\delta^2 h^d$. The elementary Cauchy-Schwarz calculation

$$\begin{aligned} \|P - Q\|_{\text{TV}} &= \frac{1}{2} \int |\sqrt{dP} - \sqrt{dQ}|(\sqrt{dP} + \sqrt{dQ}) \\ &\leq \sqrt{2\{1 - \rho(P, Q)\}} \end{aligned}$$

then gives

$$\|P_\omega^{\otimes n} - P_{\omega^{(j)}}^{\otimes n}\|_{\text{TV}} \leq C\sqrt{n\delta^2 h^d}.$$

Choose

$$h = a_0 n^{-1/(2\gamma+d)},$$

where the fixed $a_0 > 0$ is small enough that the last upper bound is at most $1/2$. For all sufficiently large n , this h is also within the range of the packing lemma. Thus every adjacent pair has testing affinity

$$1 - \|P_\omega^{\otimes n} - P_{\omega^{(j)}}^{\otimes n}\|_{\text{TV}} \geq \frac{1}{2}.$$

It remains to convert testing difficulty into the weighted set loss. Let $L_{\omega j}(H)$ be the part of $d_H(H, G_{P_\omega}; P_\omega)$ on the j -th disjoint packing patch. The packing lemma states that for every measurable H ,

$$L_{\omega j}(H) + L_{\omega^{(j)} j}(H) \geq s_j, \quad s_j \geq c_1 \delta^2 h^{d-1}.$$

For any set estimator $\hat{G}_n$, integration with respect to the common part of two adjacent sample laws gives

$$\mathbb{E}_{P_\omega^{\otimes n}} L_{\omega j}(\hat{G}_n) + \mathbb{E}_{P_{\omega^{(j)}}^{\otimes n}} L_{\omega^{(j)} j}(\hat{G}_n) \geq s_j \{1 - \|P_\omega^{\otimes n} - P_{\omega^{(j)}}^{\otimes n}\|_{\text{TV}}\} \geq \frac{s_j}{2}.$$

Indeed, on a dominating measure with densities p, q , the integral over $\min\{p, q\}$ is at least $s_j \int \min\{p, q\} = s_j(1 - \|P - Q\|_{\text{TV}})$.

Average the risk uniformly over ω , sum over the disjoint patches, and pair the vertices along each coordinate. Each unordered edge is counted twice in the uniform vertex sum, so

$$\begin{aligned} \sup_{P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)} \mathbb{E}_P[d_H(\hat{G}_n, G_P; P)] &\geq 2^{-M} \sum_{\omega} \sum_{j=1}^M \mathbb{E}_{P_\omega^{\otimes n}} L_{\omega j}(\hat{G}_n) \\ &\geq cM\delta^2 h^{d-1}. \end{aligned}$$

For $d \geq 2$, the packing lemma gives $M \geq c_0 h^{-(d-1)}$; for $d = 1$, it gives $M = 1$, and the same conclusion holds with $h^{d-1} = 1$. Hence every measurable estimator satisfies

$$\sup_{P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)} \mathbb{E}_P[d_H(\hat{G}_n, G_P; P)] \geq c\delta^2 = ch^{2\gamma} = cn^{-2\gamma/(2\gamma+d)}.$$

Taking the infimum over estimators and using $\mathcal{M}_{\text{tr}}(\gamma; \kappa) = \mathcal{M}(1, \gamma; \kappa)$ proves the asserted minimax lower bound. $\square$

Lemma 5 (lem:calibrated-direct-regression-upper). *For every fixed calibrated κ , every $\gamma > d$, and all sufficiently large n , there is one explicit armwise local-polynomial set estimator, independent of the unknown member law, for which*

$$\sup_{P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)} \mathbb{E}_P \left[d_H(\hat{G}_n, G_P; P) \right] \leq C n^{-2 \min\{\beta, \gamma\} / \{2 \min\{\beta, \gamma\} + d\}}.$$

Consequently,

$$\mathfrak{R}_n(1, \gamma; \kappa) \leq C n^{-2 \min\{\beta, \gamma\} / \{2 \min\{\beta, \gamma\} + d\}}.$$

Proof. Fix the calibrated κ , let $P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)$, and put

$$s = \min\{\beta, \gamma\}.$$

We first show that the local CATE restriction is global in the calibrated class. At $t = c$, $\text{ass} : \text{lower} - \text{margin}$ and $c_m = c^{-1}$ give

$$1 \leq P_X\{0 < |\tau_P(X)| \leq c\} \leq 1.$$

Consequently $0 < |\tau_P(X)| \leq c$ almost surely. Since τ_P is continuous, $f_P = 2^{-d}$ almost everywhere by $\text{lem} : \text{uniform} - \text{design}$, and every nonempty relatively open subset of the cube has positive Lebesgue measure, it follows that $|\tau_P(x)| \leq c$ for every $x \in \mathcal{X}$. Hence $\mathcal{D}_P(c) = \mathcal{X}$, and $\text{ass} : \text{near} - \text{band} - \text{holder}$ yields $\tau_P \in \mathbf{H}_\gamma(1; \mathcal{X})$. Together with $\mu_0 \in \mathbf{H}_\beta(1; \mathcal{X})$ and $\mu_1 = \mu_0 + \tau_P$ almost everywhere, the standard representative of each arm regression belongs to $\mathbf{H}_s(C_H; \mathcal{X})$, where C_H depends only on the fixed class parameters. This last assertion follows directly from the Taylor-remainder definition: a function of higher Hölder order on the fixed cube satisfies the lower-order remainder bound after enlarging the radius by a constant depending only on the two orders and the cube diameter, and Hölder balls are closed under addition.

We now give one estimator, common to every member law. Set

$$h = n^{-1/(2s+d)}, \quad p_s = \lceil s \rceil - 1,$$

and use the window, kernel, and shifted-Legendre basis from $\text{def} : \text{window} - \text{basis}$ with order p_s . Write $r_{x,h}(u) = r_{x,h,p_s}(u)$ and $r_x = r_{x,h,p_s}(x)$. For $a \in \{0, 1\}$, define

$$\begin{aligned} \tilde{Q}_a(x) &= \frac{1}{n} \sum_{i=1}^n K_{x,h}(X_i) r_{x,h}(X_i) r_{x,h}(X_i)^\top \mathbf{1}\{A_i = a\}, \\ \tilde{R}_a(x) &= \frac{1}{n} \sum_{i=1}^n K_{x,h}(X_i) r_{x,h}(X_i) \mathbf{1}\{A_i = a\} Y_i. \end{aligned}$$

Let $\lambda = 2^{-d-1}c_\pi$ and $\Lambda = 2^{1-d}$. For a symmetric matrix $M = V \text{diag}(\lambda_j) V^\top$, define $T_{[\lambda, \Lambda]}(M) = V \text{diag}(\min\{\Lambda, \max\{\lambda, \lambda_j\}\}) V^\top$. Set

$$\hat{Q}_a(x) = T_{[\lambda, \Lambda]} \{\tilde{Q}_a(x)\}, \quad \hat{\mu}_a(x) = \text{clip}_{[-1, 1]} \{r_x^\top \hat{Q}_a(x)^{-1} \tilde{R}_a(x)\}.$$

Finally, put

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x), \quad \hat{G}_n = \{x \in \mathcal{X} : \hat{\tau}(x) > 0\}.$$

All operations are Borel measurable, and the construction uses only the sample and the fixed class parameters, not the unknown member law.

We establish a uniform pointwise exponential bound. Let

$$\zeta_a(u) = P(A = a \mid X = u),$$

so strict overlap gives $c_\pi \leq \zeta_a(u) \leq 1 - c_\pi$. Define the population quantities

$$Q_a(x) = \mathbb{E}_P[K_{x,h}(X)r_{x,h}(X)r_{x,h}(X)^\top \mathbf{1}\{A = a\}], \quad R_a(x) = \mathbb{E}_P[K_{x,h}(X)r_{x,h}(X)\mathbf{1}\{A = a\}Y].$$

By the uniform density, the change of variables $z = T_{x,h}(u)$, and orthonormality of the finite basis,

$$2^{-d}c_\pi I \preceq Q_a(x) \preceq 2^{-d}(1 - c_\pi)I.$$

Thus the spectrum of $Q_a(x)$ lies strictly inside $[\lambda, \Lambda]$, uniformly in P , x , and h .

Let $q_{a,x}$ be the degree- p_s Taylor polynomial of μ_a at x , expressed in the basis as $q_{a,x}(u) = v_{a,x}^\top r_{x,h}(u)$. The Taylor-remainder condition and the diameter of $W_{x,h}$ give

$$\sup_{u \in W_{x,h}} |\mu_a(u) - q_{a,x}(u)| \leq Ch^s.$$

Since $R_a(x) = \mathbb{E}[K_{x,h}(X)r_{x,h}(X)\zeta_a(X)\mu_a(X)]$, the identity defining Q_a yields

$$\begin{aligned} & |r_x^\top Q_a(x)^{-1} R_a(x) - \mu_a(x)| \\ & \leq |q_{a,x}(x) - \mu_a(x)| + |r_x^\top Q_a(x)^{-1} \mathbb{E}[K_{x,h}(X)r_{x,h}(X)\zeta_a(X)\{\mu_a(X) - q_{a,x}(X)\}]| \\ & \leq Ch^s. \end{aligned}$$

Here all basis norms and inverse norms are bounded by constants because p_s and d are fixed.

For completeness, we derive the stochastic bound used next. Each centered entry of $\tilde{Q}_a(x) - Q_a(x)$ and $\tilde{R}_a(x) - R_a(x)$ is an average of independent variables bounded in absolute value by Ch^{-d} , with second moment at most Ch^{-d} . The bounds follow from $V_{x,h} \geq 2^{-d}h^d$, boundedness of the fixed-degree basis, and $|Y| \leq 1$. For any centered summand W satisfying these two bounds, expansion of its exponential and the inequality $m! \geq 2 \cdot 3^{m-2}$ for $m \geq 2$ give, whenever $|\lambda|Ch^{-d} < 3$,

$$\mathbb{E}e^{\lambda W} \leq \exp\left\{\frac{C\lambda^2 h^{-d}}{2(1 - |\lambda|Ch^{-d}/3)}\right\}.$$

Exponential Markov applied to the independent sum and optimization in λ , followed by a union bound over the fixed number of vector and matrix entries, therefore gives

$$P\left(\|\tilde{Q}_a(x) - Q_a(x)\|_F + \|\tilde{R}_a(x) - R_a(x)\|_2 > t\right) \leq C_1 \exp\{-C_2 nh^d \min(t^2, t)\}.$$

The constants are uniform over P , x , and a .

Because $Q_a(x)$ has spectrum in the clipping interval, orthogonal diagonalization and scalar projection onto $[\lambda, \Lambda]$ give

$$\|\hat{Q}_a(x) - Q_a(x)\|_F \leq \|\tilde{Q}_a(x) - Q_a(x)\|_F, \quad \|\hat{Q}_a(x)^{-1}\|_{\text{op}} \leq \lambda^{-1}.$$

The resolvent identity, the deterministic boundedness of $R_a(x)$, and clipping of the scalar output, which cannot increase its distance from $\mu_a(x) \in [-1, 1]$, now imply constants $B_0, B_1, B_2 > 0$ such that

$$\sup_{P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)} \sup_{x \in \mathcal{X}} P\{|\hat{\tau}(x) - \tau_P(x)| > B_0 h^s + t\} \leq B_1 \exp\{-B_2 nh^d \min(t^2, t)\}.$$

Since $|\tau_P(x)| \leq c < 1/4$, only bounded t is needed below, and the quadratic part of this exponent applies.

Put $r = h^s = n^{-s/(2s+d)}$; then $nh^d r^2 = 1$. Let $\Delta(x) = |\tau_P(x)|$. A thresholding error at x implies $\Delta(x) \leq |\hat{\tau}(x) - \tau_P(x)|$. Fubini's theorem therefore gives

$$\mathbb{E}_P[d_H(\hat{G}_n, G_P; P)] \leq \int_{\mathcal{X}} \Delta(x) P\{|\hat{\tau}(x) - \tau_P(x)| \geq \Delta(x)\} dP_X(x).$$

Set $u_0 = 2B_0r$ and $u_j = 2^j u_0$. For all sufficiently large n , $u_0 < c$. The calibrated upper-margin bound extends to every $t > 0$: it is assumed for $t \leq c$, while for $t > c$ it follows from $P_X\{0 < \Delta(X) \leq t\} \leq 1 \leq t/c = C_m t$. Thus the part with $0 < \Delta \leq u_0$ is at most

$$u_0 P_X\{0 < \Delta(X) \leq u_0\} \leq C_m u_0^2 \leq Cr^2.$$

On the shell $u_j < \Delta \leq 2u_j$, the pointwise tail bound and $\Delta - B_0r \geq \Delta/2$ give a contribution at most

$$2u_j P_X\{0 < \Delta(X) \leq 2u_j\} B_1 \exp\{-B'_2 u_j^2 / r^2\} \leq C u_j^2 e^{-c_0 4^j}.$$

Summing the shells, including only those intersecting $\{\Delta \leq c\}$, yields

$$\sup_{P \in \mathcal{M}_{\text{tr}}(\gamma; \kappa)} \mathbb{E}_P[d_H(\hat{G}_n, G_P; P)] \leq Cr^2 \sum_{j \geq 0} 4^j e^{-c_0 4^j} + Cr^2 \leq Cn^{-2s/(2s+d)}.$$

Since $s = \min\{\beta, \gamma\}$, this is the first displayed rate. Finally $\mathcal{M}_{\text{tr}}(\gamma; \kappa) = \mathcal{M}(1, \gamma; \kappa)$ by def : transverse – model, and the minimax infimum in def : minimax – risk is no larger than the risk of this single measurable estimator. This proves the consequence for $\mathfrak{R}_n(1, \gamma; \kappa)$. $\square$

Lemma 6 (lem:bkk-level-set-upper-cited). *For the higher-order local-polynomial R-learner under the smoothness model, margin condition, and matrix, density-ratio, approximation, boundedness, and sample-splitting conditions stated in Bonvini–Kennedy–Keele, the thresholded estimator satisfies $\mathbb{E}[d_H(\hat{G}_n, G_P; P)] \leq C(r_n^*)^{1+\xi}$, with the two branches in def : rate and without a condition $\xi\gamma \leq d$.*

Justification. This cited result accurately records the published conditional upper theorem. *Gap.* Its hypotheses are not certified here for the exact displayed estimator. *Consumer.* It remains a standalone background comparator and is not used by any surviving rate claim.

Theorem 4 (thm:calibrated-frontier-feasibility). *For fixed calibrated κ , $\mathcal{F}(\kappa) = \{(1, \gamma) : \gamma > d\}$. Hence every $(\xi, \gamma) \in \mathcal{F}(\kappa)$ satisfies $\xi = 1$, $\gamma > d$, and $\mathcal{M}(\xi, \gamma; \kappa) = \mathcal{M}_{\text{tr}}(\gamma; \kappa)$.*

Proof. Fix the calibrated value of κ . We prove the two set inclusions. First let $(\xi, \gamma) \in \mathcal{F}(\kappa)$. By the definition of the feasible frontier,

$$\xi\gamma > d \quad \text{and} \quad \mathcal{M}(\xi, \gamma; \kappa) \neq \emptyset.$$

Choose $P \in \mathcal{M}(\xi, \gamma; \kappa)$. The calibrated margin-compatibility lemma applies to this member and yields

$$\xi = 1 \quad \text{and} \quad \gamma > d.$$

Consequently,

$$\mathcal{F}(\kappa) \subseteq \{(1, \gamma) : \gamma > d\}. \tag{1}$$

Conversely, fix $\gamma > d$. By the calibrated-witness theorem, the observed marginal induced by P^{tr} belongs to $\mathcal{M}_{\text{tr}}(\gamma; \kappa)$. By the definition of the transverse model,

$$\mathcal{M}_{\text{tr}}(\gamma; \kappa) = \mathcal{M}(1, \gamma; \kappa).$$

It follows that $\mathcal{M}(1, \gamma; \kappa) \neq \emptyset$. Since also $1 \cdot \gamma > d$, the definition of $\mathcal{F}(\kappa)$ gives $(1, \gamma) \in \mathcal{F}(\kappa)$. Thus

$$\{(1, \gamma) : \gamma > d\} \subseteq \mathcal{F}(\kappa). \tag{2}$$

Combining (1) and (2) proves

$$\mathcal{F}(\kappa) = \{(1, \gamma) : \gamma > d\}.$$

Finally, if $(\xi, \gamma) \in \mathcal{F}(\kappa)$, the equality just proved gives $\xi = 1$ and $\gamma > d$. Substitution into the definition of the transverse model then gives

$$\mathcal{M}(\xi, \gamma; \kappa) = \mathcal{M}(1, \gamma; \kappa) = \mathcal{M}_{\text{tr}}(\gamma; \kappa),$$

which proves the remaining assertion. $\square$

Justification. This theorem determines the nonempty super-margin parameter frontier before any minimax rate is discussed. *Gap.* It is a feasibility theorem and makes no statistical rate claim. *Consumer.* All calibrated super-margin risk statements may restrict without loss to $\xi = 1$ and $\gamma > d$.

Proposition 1 (prop:calibrated-direct-rate-bracket). *For every fixed calibrated κ , every $\gamma > d$, and all sufficiently large n , there are constants $0 < c \leq C < \infty$, independent of n , such that*

$$cn^{-2\gamma/(2\gamma+d)} \leq \mathfrak{R}_n(1, \gamma; \kappa) \leq Cn^{-2 \min\{\beta, \gamma\}/\{2 \min\{\beta, \gamma\}+d\}}.$$

If $\beta \geq \gamma$, the two endpoints coincide, the high-smoothness branch of r_n^ applies, and*

$$\mathfrak{R}_n(1, \gamma; \kappa) \asymp (r_n^*)^2.$$

Proof. Fix a calibrated κ and $\gamma > d$. By lem : calibrated – direct – assouad – lower, there are constants $c_{\text{lb}} > 0$ and $N_{\text{lb}} < \infty$, independent of n , such that

$$\mathfrak{R}_n(1, \gamma; \kappa) \geq c_{\text{lb}} n^{-2\gamma/(2\gamma+d)}$$

for every $n \geq N_{\text{lb}}$. By lem : calibrated – direct – regression – upper, there are constants $C_{\text{ub}} < \infty$ and $N_{\text{ub}} < \infty$, also independent of n , such that

$$\mathfrak{R}_n(1, \gamma; \kappa) \leq C_{\text{ub}} n^{-2 \min\{\beta, \gamma\}/\{2 \min\{\beta, \gamma\}+d\}}$$

for every $n \geq N_{\text{ub}}$. Set

$$N_0 = \max\{N_{\text{lb}}, N_{\text{ub}}\}, \quad c = c_{\text{lb}}, \quad C = \max\{c_{\text{lb}}, C_{\text{ub}}\}.$$

Then $0 < c \leq C < \infty$, and both preceding inequalities hold simultaneously for every $n \geq N_0$. This proves the displayed two-sided bracket.

Suppose now that $\beta \geq \gamma$. Then $\min\{\beta, \gamma\} = \gamma$, so the lower and upper exponents in the bracket both equal $2\gamma/(2\gamma+d)$. By def : average – smoothness,

$$\bar{s} = \frac{\alpha + \beta}{2} \geq \frac{\beta}{2} \geq \frac{\gamma}{2}.$$

Since $\gamma > d$,

$$\frac{d}{4\{1 + d/(2\gamma)\}} < \frac{d}{4} < \frac{\gamma}{2} \leq \bar{s}.$$

Consequently the high-smoothness branch in def : rate applies, and

$$r_n^* = n^{-\gamma/(2\gamma+d)}.$$

Substitution into the two-sided bracket gives

$$c(r_n^*)^2 \leq \mathfrak{R}_n(1, \gamma; \kappa) \leq C(r_n^*)^2$$

for every $n \geq N_0$. The constants are positive, finite, and independent of n ; hence $\mathfrak{R}_n(1, \gamma; \kappa) \asymp (r_n^*)^2$, as claimed. $\square$

Justification. The proposition combines the explicit direct-regression achievability result with the exact-margin Assouad converse. *Gap.* Its exponents coincide when $\beta \geq \gamma$ and can differ when $\beta < \gamma$. *Consumer.* The residual OEQ and open-obligation record use this proposition to separate the matched subregime from the unresolved $\beta < \gamma$ bracket.

7 Interpretation

The equality $\mathcal{F}(\kappa) = \{(1, \gamma) : \gamma > d\}$ is induced by the fixed calibration $c_m = C_m = c^{-1}$, together with interior crossing and Hölder regularity. It should not be read as a universal margin-smoothness frontier. Within this calibrated slice, transverse graph perturbations preserve the exact margin and yield the oracle Assouad lower rate; nuisance-sensitive indistinguishability under known uniform design is the unresolved step.

Technical internal limitation (diagnostic only)

For $\beta < \gamma$, no proved construction in the frozen core yields a lower bound of order $(r_n^*)^2$. In the low branch, the natural fuzzy family has transition-region divergence of order $nh^d\delta^2$, which diverges at $\delta = r_n^*$. The current direct upper bound is also slower than the benchmark when $\beta < \gamma$. Therefore neither a two-branch match nor sharpness of the remaining bracket is claimed.

8 Honest scope

All feasibility statements are confined to the fixed calibrated cube. The exact minimax conclusion is confined to $\beta \geq \gamma$, where the direct lower and upper exponents coincide and the high-smoothness branch applies. For $\beta < \gamma$, the note reports the proved bracket and the precise uniform-density obstruction; it does not claim that the Bonvini–Kennedy–Keele upper transfer is rate-sharp or that the Kennedy–Balakrishnan–Robins–Wasserman lower construction belongs to this law class.

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